

PATIALA, India

WMO: 421010

Lat: 30.3578N

Lon: 76.4499E

Elev: 823

StdP: 14.26

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 41.4	(c) 43.4	(d) 37.5	(e) 33.9	(f) 55.8	(g) 39.7	(h) 37.1	(i) 53.8	(j) 9.8	(k) 58.4	(l) 7.8	(m) 59.7	(n) 0.9	(o) 320	(p) 0.344

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 18.9	(c) 107.2	(d) 76.2	(e) 104.6	(f) 75.9	(g) 101.7	(h) 76.1	(i) 85.1	(j) 93.0	(k) 84.5	(l) 92.7	(m) 83.7	(n) 91.8	(o) 3.5	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
83.4	180.0	89.2	82.6	175.5	88.9	81.9	171.2	88.4	50.7	94.0	49.9	93.9	49.0	91.7	90.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.6	(b) 7.0	(c) 5.5	(e) 37.6	(f) 111.5	(g) 3.0	(h) 1.8	(i) 35.4	(j) 112.8	(k) 33.7	(l) 113.9	(m) 32.0	(n) 114.9	(o) 29.9	(p) 116.2		
(a) 8.6	(b) 7.0	(c) 5.5	(e) 37.4	(f) 86.2	(g) 3.0	(h) 1.1	(i) 35.3	(j) 87.0	(k) 33.5	(l) 87.7	(m) 31.8	(n) 88.4	(o) 29.5	(p) 89.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	75.4	54.5	61.0	70.4	81.9	88.8	89.6	86.6	85.2	82.9	77.2	67.7	58.2
	DBStd	12.75	4.21	4.40	5.54	5.36	4.80	5.04	3.42	2.86	3.00	3.99	3.92	4.40
	HDD50	13	11	0	0	0	0	0	0	0	0	0	0	2
	HDD65	695	324	125	15	0	0	0	0	0	0	0	19	212
	CDD50	9284	152	309	632	958	1201	1187	1135	1091	988	844	530	257
	CDD65	4491	0	14	182	508	736	737	670	626	538	379	99	2
	CDH74	58331	2	100	1708	6586	11101	11197	9251	8090	6128	3418	730	20
CDH80	31791	0	7	621	3912	7158	7094	4918	3818	2684	1456	123	0	
Wind	WSAvg	1.6	1.5	2.1	2.1	2.2	2.4	2.1	1.7	1.3	1.3	0.8	0.9	1.2
Precipitation	PrecAvg	23.6	0.7	1.0	0.7	0.6	0.9	3.4	6.1	5.7	3.7	0.4	0.1	0.4
	PrecMax	46.5	2.7	3.0	2.9	3.0	6.7	18.1	21.1	19.3	10.7	3.5	0.8	2.3
	PrecMin	12.0	0.0	0.0	0.0	0.0	0.0	0.3	0.4	0.0	0.0	0.0	0.0	0.0
	PrecStd	8.3	0.8	0.8	0.8	0.7	1.3	3.6	4.6	4.7	3.3	0.9	0.2	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.3	80.9	94.2	105.9	110.6	111.1	100.5	96.2	94.5	93.3	85.9	76.7
		MCWB	60.8	67.3	72.4	74.6	77.6	77.6	82.8	82.6	80.8	76.1	67.5	61.9
	2%	DB	68.7	76.9	89.9	102.6	107.6	107.5	97.5	94.3	93.0	91.3	83.3	73.3
		MCWB	58.8	64.8	70.5	73.7	75.1	77.8	82.9	83.0	80.7	76.0	66.5	60.7
	5%	DB	66.4	74.0	86.4	99.9	105.2	104.8	95.7	92.9	91.8	89.7	81.0	70.8
		MCWB	57.4	63.0	69.9	72.4	74.8	77.8	82.9	82.8	80.7	75.2	65.6	59.9
	10%	DB	64.0	71.2	83.0	96.9	102.5	101.4	93.8	91.6	90.5	87.8	78.5	68.3
		MCWB	56.1	61.0	68.3	71.5	75.4	78.4	82.8	82.6	80.4	74.5	64.7	58.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.2	69.2	76.8	83.5	85.1	85.4	85.5	84.9	81.9	71.7	65.8	
		MCDB	69.0	77.6	89.6	100.5	101.5	95.3	92.6	91.9	91.5	89.3	81.0	72.7
	2%	WB	60.5	66.6	73.8	79.2	83.1	84.4	85.1	84.8	83.6	79.6	69.9	63.1
		MCDB	67.0	74.5	85.3	96.9	100.5	96.1	92.3	91.1	90.1	87.6	79.2	69.9
	5%	WB	58.6	64.4	71.4	76.4	80.6	83.3	84.5	84.0	82.5	78.0	68.0	61.3
		MCDB	64.4	72.1	82.6	94.4	97.6	95.0	91.8	90.1	88.8	86.1	77.9	68.2
	10%	WB	56.9	62.4	69.3	73.8	78.4	82.2	83.7	83.3	81.5	76.3	66.1	59.6
		MCDB	62.3	69.3	80.2	91.7	95.7	93.5	90.9	89.3	87.7	84.5	75.5	66.6
Mean Daily Temperature Range	5% DB	MDBR	17.6	20.2	22.8	27.3	24.5	18.9	11.5	11.1	14.5	22.3	24.6	20.5
		MCDBR	22.0	23.1	26.2	30.1	28.2	25.0	14.3	13.0	16.2	22.8	26.3	23.9
		MCWBR	12.6	11.9	10.2	10.7	9.5	8.2	5.0	5.2	6.6	9.3	11.2	12.5
	5% WB	MCDBR	19.3	21.2	24.2	26.9	24.9	17.9	12.6	11.9	14.2	19.7	23.6	21.8
		MCWBR	12.3	12.4	11.2	13.6	11.9	7.6	5.4	5.7	7.0	9.6	11.8	12.8
Clear-Sky Solar Irradiance	taub		0.474	0.477	0.495	0.565	0.668	0.747	0.758	0.643	0.567	0.636	0.582	0.515
	taud		1.892	1.935	1.928	1.804	1.634	1.538	1.544	1.771	1.888	1.636	1.693	1.802
	Ebn at Noon		229	245	251	239	215	198	195	217	228	196	193	206
	Edh at Noon		52	55	59	69	82	90	89	70	60	73	63	55
All-Sky Solar Radiation	RadAvg		983	1338	1785	2091	2279	2042	1704	1652	1682	1582	1268	1035
	RadStd		97	125	96	89	78	156	134	108	112	83	60	68

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page